Variable Selection for Wasserstein Robust Hypothesis Testing

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Abstract

Variable selection for Wasserstein robust testing admits an exact finite formulation and a maximum-affine dual representation. Dual couplings generate global cuts. The analysis includes certificates for inexact solves and data-dependent bounds on the evaluated selections.

1 Problem formulation

Let $[D] = \{1, \dots, D\}$ and let $0 \le d \le D$ be an integer. For a measurable score $f : \mathbb{R}^D \to \mathbb{R}$, define

$$R(f) = \sup_{\substack{P_0 \in \mathcal{U}_0 \\ P_1 \in \mathcal{U}_1}} \left\{ \mathbb{E}_{P_0} \ell(-f(X)) + \mathbb{E}_{P_1} \ell(f(X)) \right\}. \tag{1}$$

The loss ℓ is finite, nonnegative, convex and nondecreasing, with $\ell(0) > 0$. Write $\ell_0(s) = \ell(-s)$ and $\ell_1(s) = \ell(s)$. For $S \subseteq [D]$, let $\mathrm{Proj}_S : \mathbb{R}^D \to \mathbb{R}^{|S|}$ be the coordinate projection and define

$$\mathcal{G}_S = \{ g : \mathbb{R}^{|S|} \to \mathbb{R} : \ g \text{ is Borel measurable} \}.$$

The variable-selection problem is

$$\boxed{V(S) = \inf_{g \in \mathcal{G}_S} R(g \circ \mathrm{Proj}_S), \qquad \mathrm{OPT} = \min_{|S| \le d} V(S).} \tag{2}$$

The N observations are $x_1, \dots, x_N \in \mathbb{R}^D$. Nonempty index sets I_0, I_1 partition $[N]$ by class, with within-class masses $q_i > 0$:

$$\sum_{i \in I_k} q_i = 1, \qquad \widehat{P}_k = \sum_{i \in I_k} q_i \delta_{x_i}, \qquad \mathcal{U}_k = \{ P : W_1(P, \widehat{P}_k) \le \theta_k \}, \quad k = 0, 1.$$

Here δ_x is unit mass at x, $\theta_k > 0$, and W_1 uses ℓ_1 cost on distributions with finite first moments. Unweighted samples have $q_i = 1/|I_k|$ for $i \in I_k$.

2 Dual reformulation

Fix S and write

$$x_i^S = \mathrm{Proj}_S x_i, \qquad \Delta_{ir,j} = |x_{ij} - x_{rj}|, \qquad c_{ir}^S = \|x_i^S - x_r^S\|_1 = \sum_{j \in S} \Delta_{ir,j}.$$

Here $i, r \in [N]$ and $j \in [D]$. A coupling $\pi_{ir} \ge 0$ transports mass from source i to target r. Define

$$\mathcal{T}_k(S) = \left\{ \pi \ge 0 : \sum_r \pi_{ir} = q_i, \quad \sum_{i \in I_k, r} \pi_{ir} c_{ir}^S \le \theta_k \right\}.$$

With target-score variables $u = (u_1, \dots, u_N)$, the finite transport representation is

$$V(S) = \inf_{u \in \mathbb{R}^N} \sum_{k=0}^{1} \max_{\pi \in \mathcal{T}_k(S)} \sum_{i \in I_k, r} \pi_{ir} \ell_k(u_r). \tag{3}$$

Dualizing each transport LP introduces source-loss bounds $a_i \in \mathbb{R}$ and transport prices $\lambda_k \ge 0$. With $J(a, \lambda) = \sum_i q_i a_i + \sum_k \theta_k \lambda_k$, this gives

$$\begin{aligned} V(S) = \min_{a, u \in \mathbb{R}^N, \ \lambda \ge 0} \quad & J(a, \lambda) \\ \text{subject to} \quad & \ell_k(u_r) \le a_i + \lambda_k c_{ir}^S, \quad i \in I_k, \ r \in [N], \ k = 0, 1. \end{aligned} \tag{4}$$

Appendix A proves exactness and recovers an optimal test.

For the binary indicator $z = \mathbf{1}_S$, replace the switched prices $\lambda_k z_j$ by coordinate prices w_{kj}. Since $a_i \geq 0$,

$$\sum_k \theta_k w_k = z \sum_k \theta_k \lambda_k \leq zJ(a,\lambda) \quad \text{when } w_k = \lambda_k z.$$

For fixed $z \in [0,1]^D$, define

$$\begin{aligned} F(z) = \min_{a,u,\lambda,w} \quad & J(a,\lambda) \\ \text{subject to} \quad & \ell_k(u_r) \leq a_i + \sum_j \Delta_{ir,j} w_{kj}, \quad i \in I_k,\ r \in [N],\ k = 0,1, \\ & 0 \leq w_k \leq \lambda_k \mathbf{1}, \qquad k = 0,1, \\ & \theta_0 w_0 + \theta_1 w_1 \leq zJ(a,\lambda), \\ & \lambda \geq 0, \qquad a,u \in \mathbb{R}^N. \end{aligned} \tag{5}$$

Here $w_k = (w_{k1}, \ldots, w_{kD})^\top$ and $\mathbf{1} = (1, \ldots, 1)^\top$; vector inequalities are coordinatewise. For binary z, under the other constraints,

$$\theta_0 w_0 + \theta_1 w_1 \leq zJ \quad \Leftrightarrow \quad w_{kj} = 0 \quad \text{for } z_j = 0,\ k = 0,1.$$

When $z_j = 1$, that particular inequality is redundant and may be omitted:

$$\theta_0 w_{0j} + \theta_1 w_{1j} \leq \sum_k \theta_k \lambda_k \leq J.$$

Deleting the entire family also removes the zero-coordinate restrictions, giving $V([D])$ for every z. Retaining it gives $F(\mathbf{1}_S) = V(S)$; Proposition B.1 proves equality of the projected feasible sets and preservation of the objective.

A dual of (5), with a scaled transport coupling $\Pi \geq 0$, a scalar $\alpha \geq 0$, and coordinate multipliers $\mu \geq 0$, is

$$\begin{aligned} F(z) = \max_{\Pi,\alpha,\mu} \quad & H(\Pi) \\ \text{subject to} \quad & (\Pi,\alpha,\mu) \in \mathcal{D}, \qquad \alpha + \mu^\top z = 1. \end{aligned} \tag{6}$$

Here

$$H(\Pi) = \sum_{r=1}^{N} \inf_{s \in \mathbb{R}} \sum_{k=0}^{1} \sum_{i \in I_k} \Pi_{ir} \ell_k(s),$$

and, with $(v)_+ = \max\{v, 0\}$,

$$\mathcal{D} = \left\{ (\Pi,\alpha,\mu) \geq 0 : \begin{array}{l} \displaystyle\sum_r \Pi_{ir} = \alpha q_i, \quad i \in [N], \\ \displaystyle\sum_j \left(\sum_{i \in I_k, r} \Pi_{ir} \Delta_{ir,j} - \theta_k \mu_j \right)_+ \leq \alpha \theta_k, \quad k = 0,1 \end{array} \right\}. \tag{7}$$

The coupling has source masses αq_i; H minimizes its weighted loss over one score per target. The Lagrangian derivation is in Proposition B.2.

For a normalized tuple $\alpha + \mu^\top z = 1$, scaling all three dual variables by $\tau \geq 0$ preserves $\mathcal{D}$ and gives

$$H(\tau\Pi) = \tau H(\Pi), \qquad \max_{\tau \geq 0} \{2\sqrt{\tau H(\Pi)} - \tau\} = H(\Pi), \quad \tau^* = H(\Pi).$$

Optimizing this scale on every ray removes the normalization:

$$\boxed{F(z) = \max_{(\Pi,\alpha,\mu) \in \mathcal{D}} \{2\sqrt{H(\Pi)} - \alpha - \mu^\top z\}.} \tag{8}$$

Proposition B.3 proves this equivalence and attainment. A normalized optimizer $(\bar{\Pi}, \bar{\alpha}, \bar{\mu})$ gives the affine optimizer $F(z)(\bar{\Pi}, \bar{\alpha}, \bar{\mu})$; the scaling must also be applied to the cut slope.

3 Cutting-plane algorithm

Let $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \le d\}$. At z^t, evaluate $F(z^t)$ by (8) and take an optimizer (Π^t, α^t, μ^t) with the bounds $\mu_j^t \le M_j$ from Proposition C.2. Its subgradient $-\mu^t$ gives

$$F(z) \ge 2\sqrt{H(\Pi^t)} - \alpha^t - (\mu^t)^\top z = F(z^t) - (\mu^t)^\top (z - z^t). \tag{9}$$

The next selection and lower bound solve

$$\begin{aligned} (z^{t+1}, \eta^{t+1}) \in \underset{z \in \mathcal{Z}_d,\ \eta \ge 0}{\arg\min} \quad & \eta \\ \text{subject to} \quad & \eta \ge F(z^s) - (\mu^s)^\top (z - z^s), \qquad s = 1, \dots, t. \end{aligned} \tag{10}$$

For exact solves, writing $U_t = \min_{s \le t} F(z^s)$ and T for the last iteration, Appendix C proves

$$\begin{aligned} & \eta^{t+1} \le \mathrm{OPT} \le U_t, \\ & 0 \le R(f^*) - \mathrm{OPT} \le U_T - \eta^{T+1} \le \varepsilon, \qquad T \le \sum_{r=0}^{d} \binom{D}{r}. \end{aligned} \tag{11}$$

For inexact solves, a feasible recovered score supplies an upper bound U_t, dual feasible cuts remain global lower bounds, and a global master bound L_t gives

$$0 \le R(f_{\mathrm{inc}}) - \mathrm{OPT} \le U_t - L_t.$$

Proposition C.1 proves the certificate and the sufficient accuracy condition $\delta + \beta \le \varepsilon$ for finite termination. Propositions C.2 and C.3 additionally give bounded slopes and a data-dependent separation bound for exact search.

Algorithm 1 Exact cutting-plane method for variable selection

Require: Problem data, budget d, tolerance $\varepsilon > 0$, and $z^1 \in \mathcal{Z}_d$.
Ensure: A subset S^*, its optimal test f^*, and a risk interval of width at most ε.

$t \leftarrow 1$, $U \leftarrow +\infty$.
while true **do**
  Solve (8) exactly at z^t; choose μ^t with $0 \le \mu_j^t \le M_j$ as in Proposition C.2.
  if $F(z^t) < U$ **then**
    $(U, z^*) \leftarrow (F(z^t), z^t)$.
  end if
  Retain the cut (9) with index t.
  Solve (10) exactly to obtain (z^{t+1}, η^{t+1}).
  if $U - \eta^{t+1} \le \varepsilon$ **then**
    $S^* \leftarrow \{j : z_j^* = 1\}$.
    Solve (4) for S^*; construct g^* by (17) and set $f^* = g^* \circ \mathrm{Proj}_{S^*}$.
    return $(S^*, f^*, [\eta^{t+1}, U])$.
  end if
  $t \leftarrow t + 1$.
end while

A Proof of the finite representation

Let $x_i \in \mathbb{R}^D$, $i \in [N]$, and let nonempty sets I_0, I_1 partition $[N]$. Let $q_i > 0$, $\sum_{i\in I_k} q_i = 1$, and $\theta_k > 0$. The loss $\ell : \mathbb{R} \to [0,\infty)$ is finite, convex and nondecreasing, with $\ell(0) > 0$; put $\ell_0(s) = \ell(-s)$ and $\ell_1(s) = \ell(s)$. For $S \subseteq [D]$, let Proj_S be coordinate projection; δ_x denotes unit mass at x. Define

$$x_i^S = \mathrm{Proj}_S\, x_i, \qquad c_{ir}^S = \|x_i^S - x_r^S\|_1, \qquad \widehat{P}_k = \sum_{i\in I_k} q_i \delta_{x_i},$$

$$\mathcal{U}_k = \{P : W_1(P, \widehat{P}_k) \le \theta_k\}, \qquad \mathcal{G}_S = \{g : \mathbb{R}^{|S|} \to \mathbb{R} :\ g \text{ is Borel measurable}\},$$

$$R(f) = \sum_{k=0}^{1} \sup_{P\in\mathcal{U}_k} \mathbb{E}_P \ell_k(f(X)), \qquad V(S) = \inf_{g\in\mathcal{G}_S} R(g \circ \mathrm{Proj}_S).$$

Here W_1 uses ℓ_1 transport cost and finite-first-moment distributions. Set

$$\begin{aligned} T(S) = \inf_{a,u\in\mathbb{R}^N,\ \lambda\in\mathbb{R}_+^2} \quad & J(a,\lambda) := \sum_i q_i a_i + \sum_{k=0}^{1} \theta_k \lambda_k \\ \text{subject to} \quad & \ell_k(u_r) \le a_i + \lambda_k c_{ir}^S, \quad i \in I_k,\ r \in [N],\ k = 0,1. \end{aligned} \tag{12}$$

$T(S)$ is the finite value; equality with $V(S)$ is proved below.

Proposition A.1. *For every $S \subseteq [D]$, $V(S) = T(S)$, the infimum defining $T(S)$ is attained, and each optimal finite solution produces a Borel score $g^* : \mathbb{R}^{|S|} \to \mathbb{R}$ with $R(g^* \circ \mathrm{Proj}_S) = V(S)$.*

Proof. Finite transport duality. Fix $u \in \mathbb{R}^N$. Define

$$\mathcal{T}_k(S) = \left\{ \pi \in \mathbb{R}_+^{|I_k|\times N} : \sum_r \pi_{ir} = q_i\ (i \in I_k), \quad \sum_{i\in I_k, r} \pi_{ir} c_{ir}^S \le \theta_k \right\}.$$

The diagonal plan is feasible and $0 \le \pi_{ir} \le q_i$; thus $\mathcal{T}_k(S)$ is nonempty and compact. With $a_i \in \mathbb{R}$, $\lambda_k \ge 0$, and indices $i \in I_k$, $r \in [N]$,

$$\begin{aligned} L_k(\pi, a, \lambda_k; u) &= \sum_{i,r} \pi_{ir} \ell_k(u_r) + \sum_i a_i \left(q_i - \sum_r \pi_{ir} \right) + \lambda_k \left(\theta_k - \sum_{i,r} \pi_{ir} c_{ir}^S \right) \\ &= \sum_i q_i a_i + \theta_k \lambda_k + \sum_{i,r} \pi_{ir} [\ell_k(u_r) - a_i - \lambda_k c_{ir}^S]. \end{aligned} \tag{13}$$

$$\sup_{\pi\ge 0} L_k = \begin{cases} \sum_i q_i a_i + \theta_k \lambda_k, & \ell_k(u_r) \le a_i + \lambda_k c_{ir}^S \quad \forall i, r, \\ +\infty, & \text{otherwise.} \end{cases}$$

LP strong duality gives

$$\begin{aligned} \max_{\pi\in\mathcal{T}_k(S)} \sum_{i\in I_k, r} \pi_{ir} \ell_k(u_r) = \min_{(a_i)_{i\in I_k},\ \lambda_k \ge 0} \quad & \sum_{i\in I_k} q_i a_i + \theta_k \lambda_k \\ \text{subject to} \quad & \ell_k(u_r) \le a_i + \lambda_k c_{ir}^S, \quad i \in I_k,\ r \in [N]. \end{aligned} \tag{14}$$

$$T(S) = \inf_{u\in\mathbb{R}^N} \sum_{k=0}^{1} \max_{\pi\in\mathcal{T}_k(S)} \sum_{i\in I_k, r} \pi_{ir} \ell_k(u_r).$$

Only the finite LP has been dualized; no minimax exchange is used.

Lower bound: $T(S) \le V(S)$. For $g \in \mathcal{G}_S$ and $\pi \in \mathcal{T}_k(S)$, choose $\tilde{x}_{ir} \in \mathbb{R}^D$ with

$$\mathrm{Proj}_S\, \tilde{x}_{ir} = x_r^S, \qquad \mathrm{Proj}_{S^c}\, \tilde{x}_{ir} = \mathrm{Proj}_{S^c}\, x_i, \qquad \|\tilde{x}_{ir} - x_i\|_1 = c_{ir}^S.$$

$$
\begin{aligned}
P_k^{\pi} &= \sum_{i\in I_k,r} \pi_{ir}\delta_{\widetilde{x}_{ir}}, \qquad W_1(P_k^{\pi},\widehat{P}_k) \le \sum_{i\in I_k,r} \pi_{ir} c_{ir}^S \le \theta_k,\\
\mathbb{E}_{P_k^{\pi}} \ell_k(g(\mathrm{Proj}_S X)) &= \sum_{i\in I_k,r} \pi_{ir}\ell_k(g(x_r^S)).
\end{aligned}
\tag{15}
$$

$$
\begin{aligned}
V(S) &\ge \inf_{g\in\mathcal{G}_S} \sum_k \max_{\pi\in\mathcal{T}_k(S)} \sum_{i\in I_k,r} \pi_{ir}\ell_k(g(x_r^S))\\
&\ge \inf_{u\in\mathbb{R}^N} \sum_k \max_{\pi\in\mathcal{T}_k(S)} \sum_{i\in I_k,r} \pi_{ir}\ell_k(u_r) = T(S).
\end{aligned}
\tag{16}
$$

Upper bound: $V(S) \le T(S)$. For any finite feasible (a,u,λ) and $y\in\mathbb{R}^{|S|}$, set

$$
\begin{gathered}
E_k(y) = \min_{i\in I_k}\{a_i + \lambda_k\|x_i^S - y\|_1\}, \qquad i_k(y) = \min\arg\min_{i\in I_k}\{a_i + \lambda_k\|x_i^S - y\|_1\},\\
r_k = \|x_{i_k}^S - y\|_1, \qquad \rho = \|x_{i_0}^S - x_{i_1}^S\|_1 \le r_0 + r_1.\\
g(y) = \begin{cases} (1-\gamma)u_{i_0} + \gamma u_{i_1}, & \gamma = \min\{1, r_0/\rho\}, \quad \rho > 0,\\ u_{i_0}, & \rho = 0. \end{cases}
\end{gathered}
\tag{17}
$$

For $\rho > 0$, $\gamma\rho \le r_0$ and $(1-\gamma)\rho \le r_1$. Convexity and finite feasibility yield

$$
\begin{aligned}
\ell_0(g(y)) &\le (1-\gamma)\ell_0(u_{i_0}) + \gamma\ell_0(u_{i_1}) \le a_{i_0} + \lambda_0\gamma\rho \le E_0(y),\\
\ell_1(g(y)) &\le (1-\gamma)\ell_1(u_{i_0}) + \gamma\ell_1(u_{i_1}) \le a_{i_1} + \lambda_1(1-\gamma)\rho \le E_1(y).
\end{aligned}
$$

For $\rho = 0$, the source–target pairs (i_0,i_0) and (i_1,i_0) give the same two bounds. The finite tie rule makes g Borel, with $|g(y)| \le \max_r |u_r|$; interpolation of the u_r is unnecessary. For $P\in\mathcal{U}_k$, disintegrate an optimal coupling as source-indexed measures ν_i (splitting repeated sources by q_i):

$$
\nu_i(\mathbb{R}^D) = q_i, \qquad \sum_{i\in I_k}\nu_i = P, \qquad \sum_{i\in I_k}\int \|x - x_i\|_1\,\nu_i(dx) \le \theta_k.
$$

$$
\begin{aligned}
\mathbb{E}_P\ell_k(g(\mathrm{Proj}_S X)) &\le \sum_{i\in I_k}\int [a_i + \lambda_k\|\mathrm{Proj}_S\, x - x_i^S\|_1]\,\nu_i(dx)\\
&\le \sum_{i\in I_k} q_i a_i + \theta_k\lambda_k.
\end{aligned}
\tag{18}
$$

$$
R(g\circ\mathrm{Proj}_S) \le J(a,\lambda), \qquad V(S) \le T(S).
$$

Attainment. The feasible point $(a,u,\lambda) = (\ell(0)\mathbf{1}, 0, 0)$ has value $C = 2\ell(0)$. Diagonal feasibility and $J \le C$ give

$$
0 \le a_i \le C/q_i, \qquad 0 \le \lambda_k \le C/\theta_k, \qquad \ell_k(u_r) \le \min_{i\in I_k}\{C/q_i + (C/\theta_k)c_{ir}^S\}.
$$

A nonconstant convex nondecreasing ℓ satisfies $\ell(t)\to\infty$ as $t\to\infty$; the two losses bound u on both sides. For constant loss take $u = 0$. A nonempty compact restriction of this closed sublevel therefore contains a minimizer; its extension attains $V(S)$. □

B Proofs of the three dual reformulations

Let I_0, I_1 partition $[N]$, with $q_i > 0$, $\sum_{i\in I_k} q_i = 1$, and $\theta_k > 0$. Let $\ell : \mathbb{R}\to[0,\infty)$ be finite, convex and nondecreasing, with $\ell(0) > 0$, $\ell_0(s) = \ell(-s)$ and $\ell_1(s) = \ell(s)$. For data $x_i\in\mathbb{R}^D$, put

$$
\Delta_{ir,j} = |x_{ij} - x_{rj}|, \qquad c_{ir}^S = \sum_{j\in S}\Delta_{ir,j}, \qquad J(a,\lambda) = \sum_i q_i a_i + \sum_k \theta_k\lambda_k.
$$

Define the finite feasible set

$$\mathcal{C}_S = \{(a,u,\lambda) : a,u \in \mathbb{R}^N,\ \lambda \geq 0,\ \ell_k(u_r) \leq a_i + \lambda_k c_{ir}^S\ (i \in I_k,\ r \in [N],\ k = 0,1)\}.$$

Set $V(S) := \min_{\mathcal{C}_S} J$. Proposition A.1 identifies this value with the original fixed-subset distributional risk. For $z \in [0,1]^D$, define

$$\begin{aligned}\mathcal{E}_z = \{(a,u,\lambda,w) :\quad & a,u \in \mathbb{R}^N,\ \lambda \geq 0, \quad 0 \leq w_{kj} \leq \lambda_k, \\ & \ell_k(u_r) \leq a_i + \sum_j \Delta_{ir,j} w_{kj}\ (i \in I_k,\ r \in [N],\ k = 0,1), \\ & \sum_k \theta_k w_{kj} \leq z_j J(a,\lambda)\ (j \in [D])\}, \qquad F(z) = \min_{\mathcal{E}_z} J.\end{aligned} \tag{19}$$

Proposition B.1 (Binary extension)**.** *For every $S \subseteq [D]$, projection onto the original variables gives*

$$\mathrm{Proj}_{(a,u,\lambda)}\, \mathcal{E}_{\mathbf{1}_S} = \mathcal{C}_S, \qquad F(\mathbf{1}_S) = V(S).$$

Both directions preserve the same objective $J(a,\lambda)$.

Proof. Let $z = \mathbf{1}_S$. For $(a,u,\lambda) \in \mathcal{C}_S$, diagonal feasibility gives $a_i \geq 0$. Set $w_k = \lambda_k z$; then

$$\ell_k(u_r) \leq a_i + \sum_j \Delta_{ir,j} w_{kj}, \qquad \sum_k \theta_k w_{kj} = z_j \sum_k \theta_k \lambda_k \leq z_j J(a,\lambda).$$

Thus $(a,u,\lambda,w) \in \mathcal{E}_z$ with the same J. Conversely, $(a,u,\lambda,w) \in \mathcal{E}_z$ gives

$$w_{kj} = 0 \quad (j \notin S), \qquad 0 \leq w_{kj} \leq \lambda_k \quad (j \in S), \qquad \ell_k(u_r) \leq a_i + \sum_{j \in S} \Delta_{ir,j} w_{kj} \leq a_i + \lambda_k c_{ir}^S.$$

Hence $(a,u,\lambda) \in \mathcal{C}_S$, again with the same J. The two inclusions prove the claim. The selector row is redundant at $z_j = 1$ and enforces $w_{kj} = 0$ at $z_j = 0$; deleting all selector rows would instead give $V([D])$ at every z. □

For $\Pi \in \mathbb{R}_+^{N \times N}$, define

$$H(\Pi) = \sum_{r=1}^{N} \inf_{s \in \mathbb{R}} \sum_{k=0}^{1} \sum_{i \in I_k} \Pi_{ir} \ell_k(s), \qquad d_{kj}(\Pi) = \sum_{i \in I_k, r} \Pi_{ir} \Delta_{ir,j}.$$

The symbol Π is a scaled transport coupling, not necessarily a probability coupling. With $(v)_+ = \max\{v, 0\}$, set

$$\mathcal{D} = \left\{(\Pi,\alpha,\mu) \geq 0 : \sum_r \Pi_{ir} = \alpha q_i\ (i \in [N]), \quad \sum_j (d_{kj}(\Pi) - \theta_k \mu_j)_+ \leq \alpha \theta_k\ (k = 0,1)\right\}. \tag{20}$$

Proposition B.2 (Normalized dual)**.** *For every $z \in [0,1]^D$, the extended minimum is attained and*

$$0 < F(z) = \max_{\substack{(\Pi,\alpha,\mu) \in \mathcal{D} \\ \alpha + \mu^\top z = 1}} H(\Pi) \leq 2\ell(0). \tag{21}$$

The normalized dual maximum is attained.

Proof. Lagrangian. Assign nonnegative multipliers $\Pi_{ir}, \beta_{kj}, \mu_j$ to the loss, price-cap and selector inequalities, respectively; keep $\lambda, w \geq 0$.

$$\begin{aligned}\mathcal{L} = J(a,\lambda) &+ \sum_{k, i \in I_k, r} \Pi_{ir} \left[\ell_k(u_r) - a_i - \sum_j \Delta_{ir,j} w_{kj}\right] \\ &+ \sum_{k,j} \beta_{kj}(w_{kj} - \lambda_k) + \sum_j \mu_j \left[\sum_k \theta_k w_{kj} - z_j J(a,\lambda)\right].\end{aligned} \tag{22}$$

Writing $\alpha = 1 - \mu^\top z$,

$$\begin{aligned}\mathcal{L} = &\sum_i a_i \left(\alpha q_i - \sum_r \Pi_{ir} \right) + \sum_{k, i \in I_k, r} \Pi_{ir} \ell_k(u_r) \\ &+ \sum_k \lambda_k \left(\alpha \theta_k - \sum_j \beta_{kj} \right) + \sum_{k,j} w_{kj} [\beta_{kj} + \theta_k \mu_j - d_{kj}(\Pi)].\end{aligned} \tag{23}$$

The infimum over free a and nonnegative λ, w is finite exactly when

$$\sum_r \Pi_{ir} = \alpha q_i, \qquad \sum_j \beta_{kj} \le \alpha \theta_k, \qquad \beta_{kj} \ge d_{kj}(\Pi) - \theta_k \mu_j. \tag{24}$$

The row sums force $\alpha \ge 0$. Elimination of $\beta \ge 0$ is exact:

$$\min_{\beta_{kj} \ge (d_{kj}(\Pi) - \theta_k \mu_j)_+} \sum_j \beta_{kj} = \sum_j (d_{kj}(\Pi) - \theta_k \mu_j)_+.$$

Minimizing over u gives $H(\Pi)$, yielding the normalized dual.

Strong duality. For $J_0 = \{j : z_j = 0\}$, feasibility forces $w_{kj} = 0$. Delete those variables and selector rows. A strict feasible point of the reduced problem is

$$u = 0, \qquad a_i = \ell(0) + 1, \qquad \lambda_k = 1, \qquad w_{kj} = z_j/2 \quad (j \notin J_0),$$
$$\ell_k(0) < a_i + \sum_j \Delta_{ir,j} w_{kj}, \qquad 0 < w_{kj} < \lambda_k, \qquad \sum_k \theta_k w_{kj} < z_j J(a, \lambda).$$

This also covers $J_0 = [D]$ after omitting w. Slater's theorem gives strong duality and dual attainment for the reduced problem. Restore the deleted-coordinate inequalities with

$$\beta_{kj} = 0, \qquad \mu_j = \max_{k=0,1} d_{kj}(\Pi)/\theta_k \quad (j \in J_0).$$

Neither normalization nor objective changes, proving full dual attainment.

Primal attainment and positivity. The feasible point $(a, u, \lambda, w) = (\ell(0)\mathbf{1}, 0, 0, 0)$ has value $C = 2\ell(0)$. On $J \le C$,

$$0 \le a_i \le C/q_i, \qquad 0 \le w_{kj} \le \lambda_k \le C/\theta_k, \qquad \ell_k(u_r) \le C/q_i + (C/\theta_k) \sum_j \Delta_{ir,j}.$$

As in the finite problem, these bounds give a compact restriction containing a minimizer (set $u = 0$ for constant loss). If its value were zero, $a = \lambda = w = 0$ would give

$$0 < 2\ell(0) \le \ell(-u_r) + \ell(u_r) \le 0,$$

a contradiction. □

Proposition B.3 (Affine dual and supporting cuts)**.** *For every $z \in [0,1]^D$,*

$$F(z) = \max_{(\Pi, \alpha, \mu) \in \mathcal{D}} \{2\sqrt{H(\Pi)} - \alpha - \mu^\top z\}. \tag{25}$$

If $(\bar{\Pi}, \bar{\alpha}, \bar{\mu})$ is a normalized dual optimizer at z, an affine dual optimizer is

$$(\Pi, \alpha, \mu) = F(z)(\bar{\Pi}, \bar{\alpha}, \bar{\mu}), \qquad H(\Pi) = F(z)^2. \tag{26}$$

Consequently F is convex. For an affine optimizer (Π^t, α^t, μ^t) at z^t,

$$-\mu^t \in \partial F(z^t), \qquad F(z) \ge F(z^t) - (\mu^t)^\top (z - z^t) \quad (z \in [0,1]^D).$$

Here the subgradient inequality is understood on the cube.

Proof. The set $\mathcal{D}$ is a cone and $H(\tau\Pi) = \tau H(\Pi) \geq 0$ for $\tau \geq 0$. For $(\Pi, \alpha, \mu) \in \mathcal{D}$, put $b = \alpha + \mu^\top z$. If $b > 0$, normalization by b gives $H(\Pi)/b \leq F(z)$, hence

$$2\sqrt{H(\Pi)} - b = \frac{H(\Pi)}{b} - \left(\sqrt{\frac{H(\Pi)}{b}} - \sqrt{b}\right)^2 \leq F(z). \tag{27}$$

If $b = 0$, nonnegativity and the row sums give $\alpha = 0$, $\Pi = 0$, so the affine value is zero. Conversely, a normalized optimizer has $H(\bar{\Pi}) = F(z) > 0$, and

$$H(F(z)\bar{\Pi}) = F(z)^2, \qquad F(z)(\bar{\alpha} + \bar{\mu}^\top z) = F(z), \qquad 2\sqrt{H(F(z)\bar{\Pi})} - F(z) = F(z).$$

Thus the affine maximum is attained at the stated rescaling. For its optimizer at z^t,

$$F(z) \geq 2\sqrt{H(\Pi^t)} - \alpha^t - (\mu^t)^\top z = F(z^t) - (\mu^t)^\top (z - z^t).$$

A supremum of affine functions is convex; this inequality gives the subgradient $-\mu^t = -F(z^t)\bar{\mu}^t$. Since H is a nonnegative infimum of linear functions, H and $\sqrt{H}$ are concave, so the fixed-z dual is a convex program. □

C Risk certificates and search cost

Let

$$\mathcal{Z}_d = \{\mathbf{1}_S : S \subseteq [D],\ |S| \leq d\}, \qquad \mathrm{OPT} = \min_{z \in \mathcal{Z}_d} F(z), \qquad F(\mathbf{1}_S) = V(S).$$

At z^s, let p_s be a primal feasible value, f_s its recovered score, and c_s a dual feasible affine cut:

$$R(f_s) \leq p_s, \qquad c_s(z) \leq F(z) \quad (z \in [0,1]^D).$$

The cut uses exact $H(\Pi)$ or a certified lower bound $0 \leq h \leq H(\Pi)$. A trial score supplies an upper bound on this infimum and cannot certify a cut.

Proposition C.1 (Certificate with inexact solves)**.** *After t evaluations define*

$$U_t = \min_{s \leq t} p_s, \qquad m_t(z) = \max\{0, c_1(z), \dots, c_t(z)\}.$$

Let (z^{t+1}, η^{t+1}) be any feasible master solution with $\eta^{t+1} \geq m_t(z^{t+1})$, and let L_t be a valid global lower bound on $\min_{z \in \mathcal{Z}_d} m_t(z)$. If f_{inc} is the score attached to the smallest p_s, then

$$L_t \leq \mathrm{OPT} \leq R(f_{\mathrm{inc}}) \leq U_t, \qquad 0 \leq R(f_{\mathrm{inc}}) - \mathrm{OPT} \leq U_t - L_t. \tag{28}$$

For $\delta, \beta \geq 0$ and $\varepsilon > 0$, if every evaluated point satisfies

$$p_s - c_s(z^s) \leq \delta, \qquad \eta^{t+1} - L_t \leq \beta, \qquad \delta + \beta \leq \varepsilon,$$

then a repeated selection $z^{t+1} = z^s$ implies $U_t - L_t \leq \varepsilon$. Thus an implementation that stops when $U_t - L_t \leq \varepsilon$ makes at most $|\mathcal{Z}_d|$ evaluations, provided these accuracies hold at each iteration.

Proof. Validity of the cuts and primal feasibility give

$$0 \leq m_t \leq F, \qquad L_t \leq \min_{\mathcal{Z}_d} m_t \leq \mathrm{OPT} \leq R(f_{\mathrm{inc}}) \leq U_t.$$

If $z^{t+1} = z^s$, then $U_t \leq p_s$ and $\eta^{t+1} \geq c_s(z^s)$, so

$$U_t - L_t \leq p_s - c_s(z^s) + \eta^{t+1} - L_t \leq \delta + \beta \leq \varepsilon.$$

Thus every nonterminal iteration selects a new element of $\mathcal{Z}_d$. The bound uses the global master bound L_t; the feasible master objective η^{t+1} alone is insufficient. □

Proposition C.2 (Data-dependent bounded slopes)**.** *Let* $\Delta_{ir,j} = |x_{ij} - x_{rj}|$, $q_i > 0$ *with* $\sum_{i\in I_k} q_i = 1$, *and* $\theta_k > 0$, *as in the extended model. Put*

$$C = 2\ell(0), \qquad A_{kj} = \sum_{i\in I_k} q_i \max_{r\in[N]} \Delta_{ir,j}, \qquad M_j = C \max_{k=0,1} \frac{A_{kj}}{\theta_k}. \tag{29}$$

At every $z \in [0,1]^D$, *there exists an affine dual optimizer with* $0 \le \mu_j \le M_j$ *for every* j. *These choices imply*

$$|F(z) - F(z')| \le \sum_j M_j |z_j - z'_j| \qquad (z, z' \in [0,1]^D).$$

Proof. For the scaled optimizer of Proposition B.3, $0 \le \bar{\alpha} \le 1$ gives $0 \le \alpha = F(z)\bar{\alpha} \le F(z) \le C$. The row sums imply $d_{kj}(\Pi) = \sum_{i\in I_{k,r}} \Pi_{ir}\Delta_{ir,j} \le \alpha A_{kj}$. Set $g_j = \max_k d_{kj}(\Pi)/\theta_k$ and $\mu'_j = \min\{\mu_j, g_j\}$. Clipped coordinates have $(d_{kj}(\Pi) - \theta_k\mu'_j)_+ = 0$; all other coordinates are unchanged. Thus $(\Pi, \alpha, \mu') \in \mathcal{D}$, and

$$F(z) \ge 2\sqrt{H(\Pi)} - \alpha - (\mu')^\top z \ge 2\sqrt{H(\Pi)} - \alpha - \mu^\top z = F(z), \qquad 0 \le \mu'_j \le g_j \le \alpha \max_k A_{kj}/\theta_k \le M_j.$$

Supporting cuts with these multipliers at z and z' give

$$-\sum_j M_j |z_j - z'_j| \le F(z) - F(z') \le \sum_j M_j |z_j - z'_j|.$$

□

Proposition C.3 (Exact search and separation)**.** *Suppose every fixed-selection problem and master problem is solved exactly. At each evaluated point choose the bounded affine optimizer of Proposition C.2, and stop when* $U_t - \eta^{t+1} \le \varepsilon$, *where* $U_t = \min_{s\le t} F(z^s)$. *For the returned score* f^* *and final iteration* T,

$$0 \le R(f^*) - \mathrm{OPT} \le U_T - \eta^{T+1} \le \varepsilon. \tag{30}$$

For every later evaluated selection S_t *and earlier selection* S_s,

$$\sum_{j\in S_t \setminus S_s} M_j > \varepsilon, \qquad s < t. \tag{31}$$

Define the weighted pseudodistance $d_M(S,T) = \sum_{j\in S\triangle T} M_j$ *and the finite packing number*

$$\mathcal{P}_\varepsilon = \max\{|\mathcal{A}| : \mathcal{A} \subseteq \{S : |S| \le d\},\ d_M(S,T) > \varepsilon \text{ for distinct } S, T \in \mathcal{A}\}.$$

Allowing $M_j = 0$,

$$T \le \mathcal{P}_\varepsilon \le \sum_{r=0}^{d} \binom{D}{r}, \qquad \#\text{oracle} = T, \quad \#\text{master} = T, \quad \#\text{recovery} = 1. \tag{32}$$

This data-dependent packing bound need not be polynomial.

Proof. In Proposition C.1, take $p_s = c_s(z^s) = F(z^s)$, $\delta = \beta = 0$, and $L_t = \eta^{t+1}$. The recovered optimum has risk U_T, proving the certificate. For $s < t$, nontermination before evaluating S_t gives

$$c_s(\mathbf{1}_{S_t}) \le \eta^t < U_{t-1} - \varepsilon \le F(\mathbf{1}_{S_s}) - \varepsilon,$$
$$\varepsilon < (\mu^s)^\top(\mathbf{1}_{S_t} - \mathbf{1}_{S_s}) = \sum_{j\in S_t\setminus S_s} \mu_j^s - \sum_{j\in S_s\setminus S_t} \mu_j^s \le \sum_{j\in S_t\setminus S_s} M_j \le d_M(S_s, S_t).$$

The evaluated selections form an ε-packing, so $T \le \mathcal{P}_\varepsilon$. Algorithm 1 gives the solve counts. Master t has D binary variables, one continuous variable and t cuts; these counts do not bound individual solve times. □